The Effect of Gamification on Students with Different Achievement Goal Orientations

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The Effect of Gamification on Students with Different Achievement Goal Orientations

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***Abstract*—In this study, we examined gamification in relation to achievement goal orientation. Achievement goal orientation is a psychological conceptualization that characterizes students' preferences to different goals, outcomes and rewards. We added achievement badges to an online learning environment used in a Data Structures and Algorithms course (N=278), and examined the responses of students with different achievement goal orientation profiles. Furthermore, we analyzed how students who were most motivated by badges differ from others in terms of achievement goal orientation and behavior.**

We found no significant differences in the behavior of the different goal orientation groups regarding badges when analyzing the log data from the learning environment. However, their attitudes towards the badges varied. On the other hand, we found that students who reported high motivation towards badges had higher mastery-intrinsic, mastery-extrinsic and performance-approach orientation, and lower avoidance-orientation than others. All of them were already high-performing before the badges were introduced. However, not all high-performing students were motivated by the badges. We also identified a small avoidance-oriented group who reported very low motivation towards the badges. The results of this paper shed light on the reasons why students respond differently to gamification and on what underlying motivational aspects might contribute to a high or low motivation towards gamification.

***Index Terms*—achievement goal orientation, gamification, achievement badges**

I. Introduction

The idea of utilizing the engaging elements of games in other settings has gained ground over the past few years. The term *gamification* has been defined as the use of game design elements in non-game contexts [1]. In educational settings, gamification has been used as an attempt to increase learners' motivation and engagement towards the learned subject [2], [3], [4]. Gamification has been shown to affect the behavior of individuals in different contexts such as marketing and education [5]. However, gamification has also received criticism for steering the focus to extrinsic rewards [6], [7]. Furthermore, we can not assume that gamification suits everyone equally well. Therefore, it is important to better understand how different learners respond to gamification in educational context.

Achievement badges are a common gamification method where a badge is awarded for successfully completing certain criteria. Typically, a badge has no real-world value but offers an instant reward in the form of a graphical icon. Badges can also be used to offer additional challenges and to guide users towards desired behavior.

Achievement goal orientation is a psychological conceptualization that characterizes students' preferences to different goals and outcomes [8]. Typically, goals are classified into *mastery goals*, which refer to an individual's preference to gain mastery of a task, and *performance goals* which refer to the preference of demonstrating competence relative to others. Goals can be further divided into *approach* and *avoidance* valences, or preference to *intrinsic* or *extrinsic* mastery goals. The goals are not mutually exclusive but each individual has a mixture of goals with varying intensities.

As badges can be seen as external rewards, we hypothesize that students with different goal orientation profiles respond differently to badges. In our previous study [9], we noticed that badges had a statistically significant impact on some aspects of students' behavior in an online learning environment (e.g. time management and avoiding mistakes). However, the group of students who were clearly affected by the badges was small. In this study, in order to understand the underlying aspects that affect students' motivation towards badges, we study gamification in the light of achievement goal orientation theory. More specifically, our research questions are:

1) Do students with different achievement goal orientation profiles respond differently to badges?
2) How do students who are motivated by the badges differ from other students in terms of goal orientation?

In this study, we added achievement badges to an online learning environment in a university-level computer science course after the first half of the course. Badges were awarded for completing exercises early, with fewer mistakes, and achieving high points. Students' achievement goal orientations were measured with a survey in the beginning of the course. The behavior of students with different achievement goal orientations was analyzed using log data collected in the learning environment. We also surveyed students' opinions about the achievement badges. Furthermore, we examined the group of students who were most and least motivated by the badges and analyzed differences in their goal orientation profiles and performance in the course.

Earlier studies related to this paper are described in Section II. Our experiment is described in Section III, and the results of the experiment are reported in Section IV. In

Section V, we present our interpretation of the results. Finally, conclusions are presented in Section VI.

II. Related work

A. Gamification

Previous research has shown that while gamification can be used to affect the behavior and attitudes of learners, not all individuals are affected in the same way. Hamari et al. [5] reviewed studies about gamification and found out that majority of the reviewed studies reported positive effects from gamification. However, they state that the effects of gamification are greatly dependent on the users and the context in which the gamification is being implemented. Furthermore, they found out that in some cases, the same aspects of gamification were considered positive by some respondents while disliked by others.

Hamari [10] studied the effects of badges in a peer-to-peer trading service and found out that those who actively monitored the badges showed increased activity in the system. However, only a relatively small portion of users showed interest towards the badges. Correspondingly, Montola et al. [11] found in their study that while some people liked badges, others did not like them or were indifferent towards them. Nevertheless, they concluded that achievement badges are a potential tool for motivating users.

In a controlled experiment on the effect of badges to students' engagement, Denny [2] reported that badges had a positive effect on the quantity of student's contributions in a learning environment where students make multiple choice questions for their peers, while the quality of the questions remained the same. He reports that while majority of the students reported that badges increased their enjoyment, perceptions towards the badges varied from positive to negative. Denny also studied differences between male and female students' perception on badges and found no significant differences, even though male students were more accustomed to playing games where badges are used.

Domínguez et al. [3] also found that students responded differently to the same gamification mechanics. For example, some students found public leaderboards motivating while some students decided to opt out from the gamified version of the learning environment because they did not like the public competition with others.

Abramovich et al. [12] studied the effect of badges within an intelligent tutoring system for teaching mathematics to middle-school students. Badges were awarded for mastering skills and participating in the system. They measured goal orientations and motivational levels in the beginning and end of the courses and found that for low-performing students, higher number of earned badges correlated with reduced performance avoidance orientation in the post-test. This could indicate that badges can be used to reduce low-performing students' concerns about performance. They found no such effect on high-performing students. Furthermore, with low-performing students, the number of earned badges was positively correlated with *performance approach* in pre-test, which could mean that students with this goal orientation are influenced by badges. In high-performing students, on the other hand, pre-test's *performance avoidance* was negatively correlated with the number of earned badges, which may indicate that students with this goal orientation are distracted by badges.

Nicholson [6] states that the concept of *situational relevance* is important in gamification. He points out that if the underlying activity is not relevant to the user, the goal of gaining gamified rewards is less likely to be relevant to that user either. However, if the challenge presented by the gamification mechanism is relevant to the user, pursuing the reward is also meaningful for the user. He also states that for some people, a simple gamification mechanism, for example a point system attached to public status, is important enough to motivate them to do a task that they have low intrinsic motivation towards. However, the same gamification mechanism might not be motivating for other people. Moreover, Nicholson defines *meaningful gamification* as: "the integration of user-centered game design elements into non-game contexts."

As reported above, in many cases the individual differences of learners affect the way gamification is perceived. The same phenomenon has also been reported in the context of serious games (e.g. [13]). However, there is not much research on whether differences in achievement goal orientation affect users' perceptions of gamification and serious games. Thus, studying these underlying motivational aspects might help us to understand why gameful methods appeal to some people while others are indifferent or negative towards them.

B. Achievement Goal Orientation

Achievement goal orientation is a psychological conceptualization that characterizes individuals' preferences to different goals, outcomes and rewards. It has been shown to predict students' motivation and performance in educational settings [14]. The concept originates from the works of Nicholls [15] and Dweck [16] who defined achievement goals as the reasons that individuals engage in achievement behavior. They identified *mastery goal* as the purpose of developing competence or task mastery and *performance goal* as the purpose of demonstrating competence relative to others. Nicholls et al. [17] later added *avoidance goals* to take into account that not all individuals are positively motivated to all tasks and try to minimize effort. Furthermore, performance goal can be divided into *performance approach* and *performance avoidance* goals which refer to the preference of demonstrating competence or avoiding to demonstrate incompetence [18]. In this study, we use an instrument by Niemivirta [19] which further divides mastery goals into *mastery-intrinsic* goals, where student aims to increase knowledge, and *mastery-extrinsic* goals where student aims at mastery but measures success with extrinsic indicators such as grades or feedback.

Our analysis uses a *person-centered approach* [20] instead of a *variable-centered approach*. In a variable-centered approach, correlations between different goal-orientation variables and educational outcomes are studied. In contrast, in a person-centered approach, individuals are clustered based on their goal orientations in order to find groups of people with similar orientation profiles. Differences between the prototypical behavior between the clusters are then studied. This approach takes into account that different configurations of goal orientations may be more meaningful than simple correlations between individual goal orientation variables and outcomes. For example, a student with both *mastery* and *performance* orientations may behave differently than a student with *mastery* but no *performance* orientation.

III. Methods

A. Design and Test Subjects

In this study, achievement badges were added to the A+ [21] online learning environment in the middle of the course. The implementation of the badge system is described by Haaranen et al. [22] in more detail. We used achievement goal orientation profiles to divide students into groups and studied differences between the groups in terms of behavior, performance and feedback. We also studied the achievement goal orientation profiles of the students who were the most or least motivated by the badges as well as the effect of badges on their behavior.

The study was conducted in the *Data Structures and Algorithms course* at *Aalto University* during Spring semester 2013. The course had online exercises that were divided into eight separate rounds. Badges were introduced in the beginning of round 5 and they stayed until the end of the course. After the course, students were asked to give feedback about the badges as part of the course feedback questionnaire.

The course had different versions for CS major and minor students. The two versions had slightly different arrangements but the online exercises were identical for both courses. However, their contribution to the final grade differed slightly, being 20% for major students and 30% for minor students.

B. Materials and Procedure

The course had a total of 54 algorithm and program visualization exercises that were divided into eight rounds. The exercises were graded on a scale from 0 to 5 where 50% of points were required to get a passing grade 1 and 90% was required to get the maximum grade 5.

On the first lecture, a pre-test was used to measure differences in the prior knowledge of the students. Then, on the first visit to the A+ learning environment, students were asked to answer a goal orientation survey by Niemivirta [19], which distinguishes five achievement goal orientations. Each goal orientation scale comprises three items that students rated using a seven point Likert scale. Composite scores were computed for each scale by averaging the scale sum scores. The survey also contained filler items and constructs that are not related to goal orientation, and those are not used in this study. Additionally, students' grade goal was asked in the survey. Students were allowed to submit an empty form. Students who did not completely fill the survey are excluded from this study. The goal orientations and examples of the items are given in Table I. The Cronbach alpha reliabilities were .90 for mastery-intrinsic orientation; .84 for mastery-extrinsic orientation; .74 for performance-approach orientation; .81 for performance-avoidance orientation; and .64 for avoidance orientation.

TABLE I
Example items from the goal orientation survey. The actual survey was in Finnish.

Orientation	Example item
Mastery-intrinsic	To acquire new knowledge is an important goal for me at school.
Mastery-extrinsic	My goal is to succeed at school.
Performance-approach	An important goal for me at school is to do better than other students.
Performance-avoidance	I try to avoid situations in which I may fail or make mistakes.
Avoidance	I try to get off with my schoolwork with as little effort as possible.

Badges were divided into three categories: *learning*, *time management*, and *carefulness*. Furthermore, each badge type had three difficulty levels: *bronze*, *silver*, and *gold*. The completion criteria for the badges are described in Table II. *Learning* and *time management* badges were given separately for each round whereas *carefulness* badges covered all the rounds of the latter half of the course.

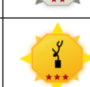
Students' activity in the A+ system was recorded, including the points and timestamps of the submissions. All the earned and available badges were presented on a separate *badge summary page*, and visits to that page were logged. Recently earned badges were also visible in the sidebar of the exercise page.

After the course, feedback was collected about the badges in the course feedback questionnaire. The questionnaire contained numerical questions about the badges with a five-point Likert scale (0=strongly disagree, 4=strongly agree) as well as an opportunity to give open ended feedback. However, the open ended feedback is excluded from this study and is analyzed in Haaranen et al. [23].

C. Analysis Methods

Students with similar goal orientation configurations were identified with Latent Profile Analysis (LPA) [24]. LPA is a probabilistic, model-based clustering method which assumes that data are generated by a mixture of underlying probability distributions and estimates model parameters with the maximum-likelihood method. The number of clusters was selected based on the Bayesian Information Criterion (BIC) with

TABLE II
Completion criteria of the badges in the three categories and difficulty levels: bronze (B), silver (S), and gold (G).

Category	Criteria	Value of X			Example icon
		B	S	G	
Time management	50% of points of the round earned at least X day(s) before the deadline	1	3	7	
Learning	Round completed with X% score	50	75	100	
Carefulness	X assignments with perfect score using only one attempt	5	10	15	

the constraint that there should be at least three clusters. With five goal orientations to study, clustering to much fewer groups would make it difficult to observe the effects of different goal orientations. The maximum BIC was achieved with four clusters. In the selected best LPA model, the covariances of the probability distributions were not allowed to vary between clusters, but volumes were allowed to vary. Clustering was performed with *MCLUST version 4 for R* [25].

The studied dependent variables that characterize students' behavior in the learning environment are: *average number of points earned per submission* which measures carefulness, *earliness* i.e. distance from submission to deadline, *earned exercise points*, *number of earned badges*, and *visits to badge summary page*. In the first half of the course, badges were not visible to students but we calculated how many badges they would have earned, in order to establish a baseline.

The difficulty of the exercise rounds in the course varies. Furthermore, the grading scheme of the course makes it impossible to reliably compare students' performance in the first half of the course (without badges) and the second half of the course (with badges) because it is possible to pass the course by completing just half of the exercises. Therefore, instead of comparing the absolute values of variables in the beginning and end of the course, we study their change. Our hypothesis is that if some group is motivated by the badges, an improvement in their behavior is observed in the latter half of the course relative to others.

To study if students with different achievement goal orientation profiles respond differently to badges, we clustered the students by their goal orientation and compared the longitudinal changes in the behavior of the groups, as well as their feedback. Next, in order to study the properties of students who were the most or least motivated by the badges, we divided the population based on their answers to the feedback question *"I found the badges motivating"*, and studied the differences between the top and bottom quartiles and the rest.

IV. Results

A total of 339 students completed at least one online exercise during the course. 82 % of them completed the achievement goal orientation survey in the beginning of the course ($N_{total} = 278$). Those who did not answer the survey, as well as students who registered for the course but did not complete any exercises are excluded from the results. In results concerning the change in performance between the first and last halves of the course, 45 students who did not complete any exercises in the latter half are excluded. Giving feedback after the course was not compulsory and therefore $N_{feedback} = 132$ students are included in the results concerning feedback. The pre-test was completed by 193 students. Because this is not a central variable in our analysis, students with missing pre-tests are not excluded from the results.

A. Achievement Goal Orientation Clusters

Students were clustered by their goal orientation values into four clusters with Latent Profile Analysis. The achievement goal orientation profiles of each group are shown in Figure 1. The groups were labeled based on the average achievement goal orientation profiles of the students in the group. We found similar groups as Tuominen-Soini et al. [8] and therefore decided to use the same labeling: *A: avoidance-oriented*, *B: mastery-oriented*, *C: indifferent*, and *D: success-oriented*. A one-way ANOVA was conducted in order to prove that the clusters differ. All goal orientation dimensions yielded significant differences with $p<0.001$ (mastery intrinsic: $F(3,274)=42.7$, mastery extrinsic: $F(3,274)=112.2$, performance approach: $F(3,274)=74.9$, performance avoidance: $F(3,274)=95.0$, and avoidance: $F(3,274)=46.2$).

The *A (avoidance-oriented)* cluster is characterized by a high avoidance orientation compared to other dimensions. The *B (mastery-oriented)* cluster has high mastery-intrinsic and mastery-extrinsic orientations. In the *C (indifferent group)*, none of the dimensions clearly dominate, while the *D (success oriented)* have high mastery-intrinsic, mastery-extrinsic, performance approach and performance avoidance orientations, but low avoidance.

In order to describe the differences of the goal orientation groups, their course grades, grade goals, performance in the beginning of the course (before badges), and their interest towards the badges are shown in Figure 2. We used violin plots to further illustrate the distributions of the values. It can be seen that the success-oriented group had the highest course grades and points from the first half of the course. They also set the grade goals higher than the others and visited the badge summary page the most. Overall, it can be seen that most of the students visit the badge summary page only few times while some students visit it multiple times.

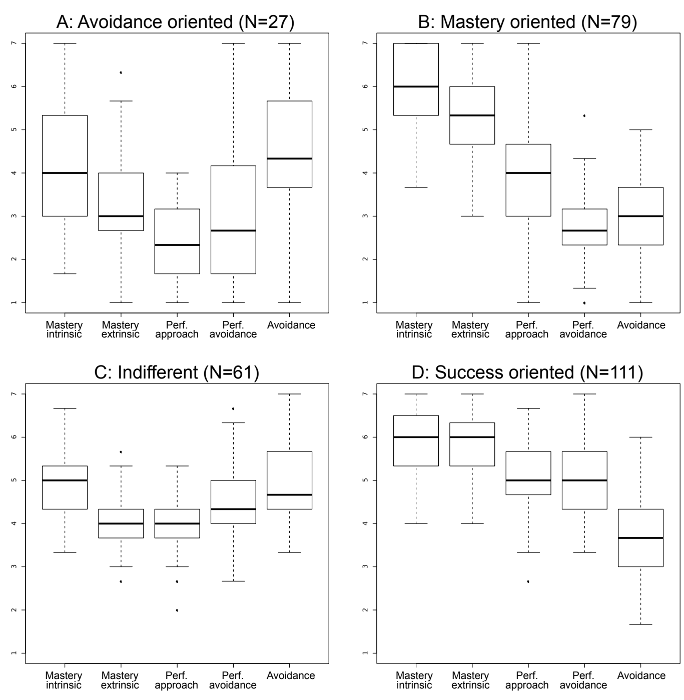


Fig. 1. Goal orientation profiles of students in different clusters.

TABLE III

DIFFERENCES IN STUDENTS' BEHAVIOR BETWEEN THE FIRST AND LAST HALVES OF THE COURSE. STUDENTS WITH NO ACTIVITY IN THE LAST HALF ARE EXCLUDED.

	A: Avoidance-oriented (N=18)		B: Mastery-oriented (N=65)		C: Indifferent (N=49)		D: Success-oriented (N=101)		ANOVA	
	M	SD	M	SD	M	SD	M	SD	F(3,229)	p-value
Increase in earned badges	-1.89	6.97	-2.20	5.89	-2.04	6.43	-2.61	6.61	0.07	0.97
Increase in points per submission	-5.52	5.74	-2.50	5.37	-4.11	5.73	-2.65	5.82	2.07	0.11
Increase in earliness	2.43	6.30	3.80	6.98	3.42	8.04	1.51	6.26	1.75	0.16
Increase in earned exercise points	-212	295	-263	243	-271	264	-301	254	0.69	0.56

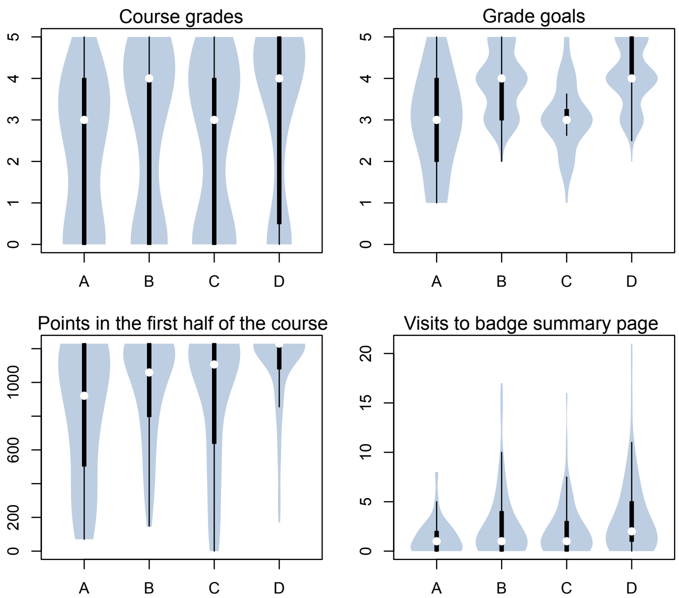


Fig. 2. Performance and other characteristics of students in different goal orientation clusters. In the violin plots, the white dot marks the median, the black bars illustrate quartiles, and the contour describes the distribution.

A one-way ANOVA was used to test for differences between the groups. The course grade differed significantly across the four groups, F(3,274)=3.156, p=0.03, as well as the grade goal, F(3,269)=26.42, p<0.01. Furthermore, points earned in the first half of the course differed significantly, F(3,274)=8.199, p<0.01. However, the pre-test score did not show significant differences, F(3,210)=2.101, p=0.1. A Kruskal-Wallis test was used to test for differences in the number of times students in different clusters visited the badge summary page. The mean number of visits in each cluster were: A=1.79, B=2.99, C=2.68, and D=3.69, but the differences were not statistically significant ($\chi^2(3)$ = 5.385, p = 0.15). The number of views was not normally distributed and therefore a non-parametric test was used.

The differences in students' behavior after adding badges to the A+ system were tested with a one-way ANOVA. The results are reported in Table III, but none of the differences were statistically significant. Only students who got at least one point for the last half of the exercises were included in the analysis to make sure that all participants had seen badges in the system. A total of 14% students did not earn any exercise points in the last half of the course (A=30%, B=16%, C=18%, and D=6%).

A Kruskal-Wallis test was used to test for differences in students' feedback between the different goal orientation clusters. The results are shown in Table V. Kruskal-Wallis was used because the scale is ordinal. Questions *"I found the badges motivating"* and *"I think that badges should be used in A+ for the next years course as well"* had statistically significantly differences between the groups. A post-hoc pairwise Wilcoxon test with Bonferroni correction (Table IV) reveals that the differences are only significant between cluster A (avoiders) and others.

B. Badge Motivation

Students were divided into quartiles based on their answer to the feedback question *"I found the badges motivating"*. These groups are labeled *BadgeTop*, *BadgeMiddle*, and *BadgeBottom*. The first and third quartile values were 1 and 3 (on a scale of 0-4), respectively, and thus students with genuinely higher or lower answers were included in the *BadgeTop* and *BadgeBottom* groups. The proportions of *BadgeTop*/*BadgeMiddle*/*BadgeBottom* students in different achievement orientation clusters are shown in Figure 3. The *BadgeTop* group only consists of students from the Mastery-oriented and Success-oriented clusters. On the other hand, students from the Avoidance-oriented cluster mainly belong to the *BadgeBottom* group.

The number of earned exercise points for students in the three groups in the first half of the course (without badges) is illustrated in Figure 4. The median points of all groups are close to maximum (1230), indicating a strong ceiling effect. Furthermore, almost all students in *BadgeTop* group have scored close to maximum points already before seeing the badges.

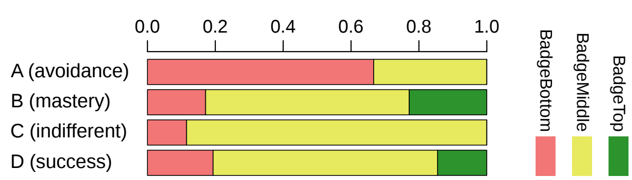


Fig. 3. Proportions of BadgeTop/BadgeMiddle/BadgeBottom groups in goal orientation clusters.

TABLE IV

P-VALUES OF THE PAIRWISE WILCOXON POST-HOC TEST WITH BONFERRONI CORRECTION FOR FEEDBACK QUESTIONS THAT DIFFERED BETWEEN THE GROUPS.

	A-B	A-C	A-D	B-C	B-D	C-D
F1	**<0.01**	**<0.01**	**<0.01**	1.00	1.00	1.00
F6	0.07	**0.03**	0.12	1.00	1.00	1.00

TABLE V
DIFFERENCES IN STUDENTS' FEEDBACK BETWEEN THE GOAL ORIENTATION CLUSTERS.

id		A: Avoidance-oriented (N=9)		B: Mastery-oriented (N=35)		C: Indifferent (N=26)		D: Success-oriented (N=62)		Kruskal-Wallis	
		M	SD	M	SD	M	SD	M	SD	$\chi^2(3)$	p-value
F1	I found the badges motivating	0.33	0.50	2.09	1.44	1.77	1.07	1.94	1.37	13.07	$<$ **0.01**
F2	Badges disturbed my work	1.11	1.54	0.71	1.23	0.62	0.98	0.81	1.05	1.60	0.66
F3	Trying to achieve badges had an effect on my behavior	0.44	1.01	1.46	1.54	1.46	1.36	1.26	1.43	4.63	0.20
F4	Visual look of the badges was good	2.33	1.22	2.89	0.96	2.73	1.04	2.89	0.98	2.70	0.44
F5	I was satisfied with the criteria for awarding badges	2.22	0.44	2.57	1.01	2.42	1.06	2.45	1.04	1.16	0.76
F6	I think that badges should be used in A+ for the next year's course as well	1.33	1.12	2.69	1.28	2.77	1.07	2.45	1.26	8.67	**0.03**

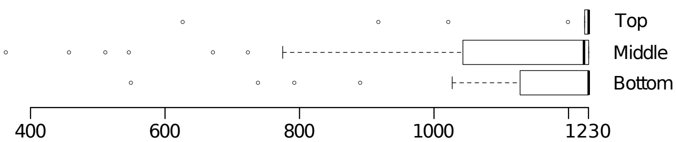


Fig. 4. Exercise points of the students in BadgeTop, BadgeMidde, and BadgeBottom groups in the first half of the course.

The achievement goal orientation profiles of the groups are shown in Figure 5. Differences in their goal orientations were tested with one-way ANOVA and the results are shown in Table VI. A Tukey HSD test was used to test the pairwise differences in cases where ANOVA showed significant differences between the groups. It can be seen that the goal orientations of the top and bottom quartiles differ significantly. The top quartile has higher *mastery-intrinsic*, *mastery-extrinsic* and *performance-approach* orientations. The top quartile also has a significantly higher *mastery-extrinsic* and lower *avoidance* orientations than the *BadgeMiddle* group.

We compared the longitudinal change in the behavior of the groups in the A+ learning environment. A series of ANOVA tests were carried out to find out if the differences are significant. The results of the tests are shown in Table VI. The top quartile had a higher increase in the *number of earned badges* and the *number of points earned per submission*. There are no significant differences in *earliness*.

V. DISCUSSION

In this study, we used a within-subject design where all students were exposed to badges in the latter half of the course. In our previous study [9], badges were used throughout the whole course in a between-subject controlled experiment. However, in a between-subject experiment, we can only measure average differences between the groups and not the effect that badges have on individual students. It is known from previous studies that typically a small group of students is motivated by badges. In a between-subject experiment, it is impossible to observe how that group would have behaved without badges, i.e. do badges improve their behavior or would they perform well regardless of badges. In this study, on the other hand, we were able to observe the behavior of the students who were motivated by the badges in the beginning of the course when badges were not yet shown.

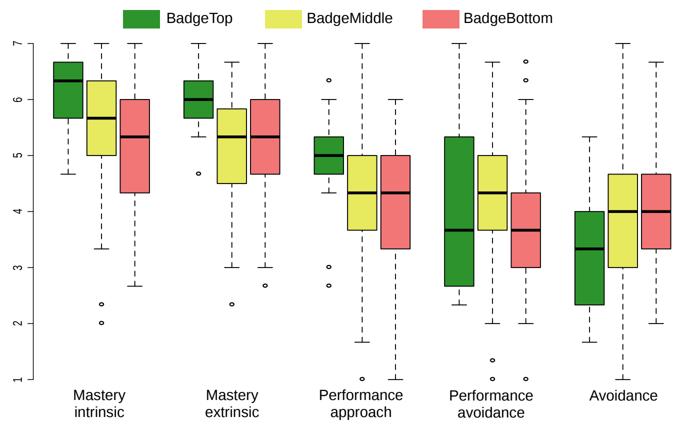


Fig. 5. Goal orientation profiles of students with different motivations towards badges.

A. Achievement Goal Orientation Clusters

To study whether students with different achievement goal orientation profiles respond differently to badges (RQ 1), students were clustered by their goal orientation profiles. Statistically significant differences were seen between the goal orientation clusters in their performance in the first half of the course, grade goals, and final grades. This indicates that achievement goal orientation is a relevant instrument when studying how individual differences affect students' behavior and performance. The results are in line with the achievement goal orientation theory as students with different goal orientations value grades differently.

We did not find differences in the increase of earned badges between the four groups (Table III). There may be several reasons for this. First, some badges were awarded for earned points, meaning that students were able to get badges by accident just by completing the exercises. Second, most students were high-performing in the beginning of the course, making it difficult to observe a significant improvement because of the ceiling effect. Finally, even though there are clear differences between the groups, the effect of badges may be small compared to other factors that affect students' motivation, such as the desire to pass the course or to earn a high grade.

Students with different goal orientation profiles had differences in their feedback answers regarding badges. Group A (avoidance-oriented) agreed significantly less to questions *"I found the badges motivating"* and *"I think that badges should*

TABLE VI
PERFORMANCE AND GOAL ORIENTATION CHARACTERISTICS OF BADGETOP, BADGEMIDDLE AND BADGEBOTTOM GROUPS. STUDENTS WITH NO ACTIVITY IN THE FIRST OR LAST HALF ARE EXCLUDED.

Variable	Top (N=17)		Middle (N=87)		Bottom (N=25)		One-way ANOVA		Tukey HSD p-values		
	M	SD	M	SD	M	SD	F(2,126)	p-value	Top-Middle	Top-Bottom	Middle-Bottom
Mastery intrinsic	18.2	2.46	16.7	3.11	15.5	3.57	3.828	**0.02**	0.17	**0.02**	0.20
Mastery extrinsic	17.9	1.95	15.4	2.81	15.6	3.59	5.530	**<0.01**	**<0.01**	**0.03**	0.93
Performance approach	14.8	2.94	12.7	3.22	11.9	4.31	3.822	**0.02**	0.05	**0.02**	0.11
Performance avoidance	12.2	4.48	12.8	3.50	11.7	4.34	0.824	0.44			
Avoidance	9.65	3.30	12.1	3.80	11.8	3.45	3.085	**0.05**	**0.04**	0.14	0.96
Increase in earned badges	1.94	3.09	-1.34	4.87	-3.56	5.92	6.361	**<0.01**	**0.03**	**<0.01**	0.11
Increase in points per submission	1.31	5.79	-2.86	5.39	-4.36	3.65	6.449	**<0.01**	**< 0.01**	**<0.01**	0.41
Increase in earliness	5.1	5.51	1.81	5.68	1.97	9.25	1.868	0.16			
Increase in earned exercise points	-166	153	-241	200	-291	219	2.021	0.14			

be used in A+ for the next years course as well" (Table IV). Other differences in feedback questions were not statistically significant. It seems that the badges did not motivate the students who prefer to minimize the effort invested in the course. This is in line with Nicholson's concept of situational relevance [6], which states that the underlying activity of gamification has to be relevant for the user. In this case, the goals of the badges were in conflict with the avoidance-oriented students' goal of minimizing effort.

It seems that based on goal-orientation alone, we can isolate a small group that is especially little motivated by badges. However, differences in goal orientation did not cause measurable differences in students' behavior related to badges. It does, however, predict overall performance to some degree.

B. Badge Motivation

To study the differences in goal orientation profiles of people with different motivation towards badges, we divided the students to three groups based on their answers to the feedback question *I found the badges motivating*. As badges can be regarded as external rewards, it is not a surprise that the *BadgeTop* group had higher *mastery extrinsic* values (Figure 5). The *BadgeTop* group also has significantly higher *mastery-intrinsic* levels even though badges are extrinsic motivators. This could simply mean that *mastery-extrinsic* and *mastery-intrinsic* orientations are strongly correlated in the population that we study.

Our badge system did not have leaderboard or other explicitly competitive elements and the system did not support sharing the badges with others. Furthermore, there were no competitive badges where only a certain number of students could earn the badge. Therefore, it is interesting that also *performance approach* varied significantly between the groups. It seems that students who prefer to compare their performance to others respond to badges even if the badges are not public or particularly competitive.

Almost all *BadgeTop* students were high-performing in the first half of the course before badges were introduced, and had scored close to full points from the exercises (Figure 4). *BadgeMiddle* and *BadgeBottom* groups were also generally high-performing but had a higher number of students who did not get as many points from the first half. It seems that all of the students who were motivated by the badges were high-performing to begin with, but not all high-performing students were motivated by the badges.

Students in the *BadgeTop* group had the highest increase in the *number of earned badges* and in *points per submission* between the first and last part of the course (Table VI). This indicates that the badges had an actual impact on some aspects of the behavior of those students who found them motivating.

The badges in our course provided additional challenges, ie. students must not only get good points from the exercises, but must solve them with few attempts in order to get the badges. These kind of badges might provide meaningful additional challenges and motivation for high-performing students who do not find enough challenge from the exercises. Low-performing students, on the other hand, may be struggling to complete the exercises in the first place, and may not welcome any additional challenge. This interpretation is in line with the findings of Abramovich et al. [12], where low-performing students responded to badges that rewarded participation but not to those that rewarded mastery. It could be that low-performing students benefit from badges that reward even small achievements and encourage to keep trying, while high-performing students are motivated by badges that provide additional challenges.

C. Validity Threats

This study is not a controlled experiment which makes it impossible to know the causal relation between badges and performance. Differences in students' performance between the first and last halves of the course might be affected by multiple reasons besides badges. For example, one group may be prone to give up more easily as exercises get more difficult.

Students with some goal orientation profile may be less likely to fill the course feedback questionnaire, which can result to a biased sample of students when studying feedback. Furthermore, students with some goal orientation may be less eager to answer the goal orientation survey in the beginning, which could cause a bias in the relative sizes of the goal orientation clusters.

Some of the deadlines of the exercise rounds were extended during the course because of various reasons. That affected especially the time management badges and might be one reason why there were no statistical differences in the change in submission times between students with different motivation towards the badges.

VI. Conclusions

In this paper, we studied the relation between achievement goal orientations and achievement badges. We found no statistically significant differences in the behavior of the different goal orientation groups regarding badges. However, their attitudes towards the badges varied. More specifically, the avoidance-oriented group reported being significantly less motivated by the badges than the others.

We examined the differences in the goal orientations of students divided into three groups based on their motivation towards the badges. We found that students who were the most motivated by the badges had significantly higher mastery intrinsic, mastery extrinsic, and performance approach orientations, and lower avoidance orientation. Furthermore, they had the highest improvement of performance after badges were introduced to the course. We also found that all of these students were high-performing already before the badges were introduced. However, not all high-performing students were motivated by the badges.

It has been found in earlier studies that typically the same gamification mechanism can be motivating to some students while disliked by others. Our results indicate that individual differences in achievement goal orientations might explain some of the differences in attitudes towards badges.

A. Future Work

Testing the hypothesis that high-performing students are motivated by additional challenge while low-performing students benefit from badges that reward any participation or progress would be an interesting topic for future research. It would also be beneficial to research how public leaderboards affect students with performance-avoidance orientation. Furthermore, when studying the effect of extrinsic motivators to intrinsic motivation or learning, performance orientations must be taken into account because any chance to compare the rewards with others may affect the behavior of performance oriented students.

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